

MBAI5310G AI Programming

Assignment 4

Decision Tree Model and Business Interpretation

ProcurePro Office Supplies

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Course: MBAI5310G AI Programming

1. Understanding the Business Problem

About the Business

ProcurePro Office Supplies is a company that helps medium sized businesses get the office products they need. Their clients include schools, clinics, consulting firms, and coworking spaces. These clients rely on ProcurePro to deliver things like printing supplies, furniture, cleaning products, and breakroom items on time. The company works with many different suppliers across Canada to make this happen.

The Problem Being Solved

The problem ProcurePro is dealing with is that not all purchase orders arrive on time. Some deliveries get delayed because of things like a supplier running out of stock, having a poor delivery history, handling too many urgent orders at once, or dealing with seasonal rushes. When a delivery is late, the client might run out of supplies, have to buy replacements at the last minute, or lose confidence in ProcurePro as a reliable partner. The business wants to spot which orders are likely to be delayed before the problem actually happens.

How Machine Learning Helps

Machine learning can help the procurement team decide which purchase orders need early follow-up. Instead of manually reviewing every single order, the team can focus their attention on the ones the model flags as high risk. Once a risky order is identified, they can reach out to the supplier early, arrange a backup option, speed up shipping, or prepare the client in advance. The model does not make the final call but it helps the team know where to look first.

Target Variable and Input Features

The target variable is `Supplier_Delay_Risk`. This column tells us whether a purchase order is high risk for a supplier delivery delay or not. Yes means high risk and No means lower risk. The input features cover several different angles of each purchase order including the supplier profile, order details, delivery setup, supplier track record, risk signals, and inventory buffer context. The `Purchase_Order_ID` column was removed before training because it is just a label and carries no predictive value.

2. Data Preparation

Dataset Overview

The dataset contains 360 purchase order records with 19 input features and one target variable. It includes a mix of categorical columns like supplier category, region, contract type, shipping mode, and payment terms, and numerical columns like order value, distance, past on time rate, and number of prior delays. Before training I cleaned the data by filling the 127 missing values in the `Backorder_History` column with the most frequently occurring value in that column since it stores category labels and mode is the right strategy for that situation.

Target Variable Distribution

The chart below shows how the orders are split between high risk and low risk. Out of 360 orders, 214 were labelled Yes and 146 were labelled No. This means the dataset is slightly imbalanced with about 59 percent of orders being high risk. This is worth keeping in mind when interpreting the model results because the model has seen more high risk examples during training.

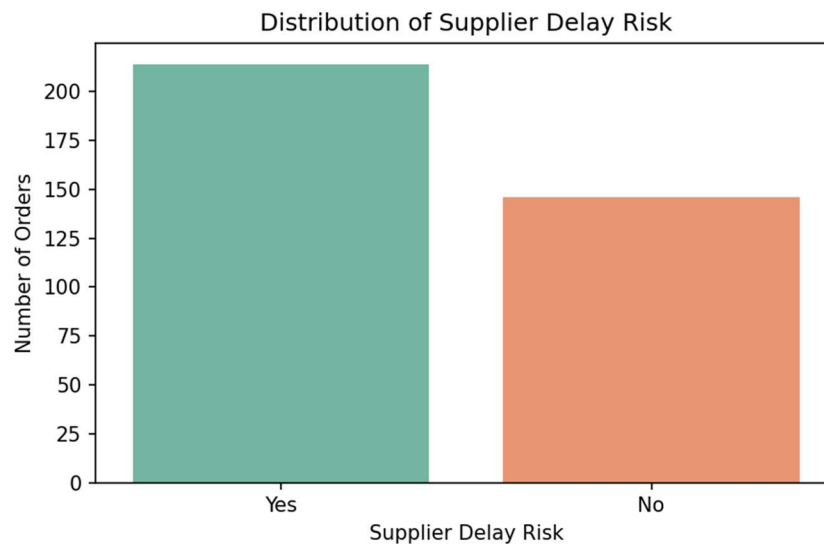


Figure 1: Distribution of the `Supplier_Delay_Risk` target variable

Encoding and Splitting

I used one hot encoding to convert all categorical columns into numerical format so the Decision Tree could work with them. This means each category value became its own binary column. Numerical columns were passed through without any changes. The data was then split into 80 percent for training and 20 percent for testing with a fixed random state to make the results reproducible.

3. Training the Decision Tree Model

Model Setup and Entropy

I chose a Decision Tree Classifier for this task because it is well suited for binary classification problems and produces results that are easy to explain in a business context. I set the maximum tree depth to 4 to keep the model simple and reduce the chance of overfitting. I also chose entropy as the splitting criterion instead of the default gini impurity.

Entropy comes from information theory and it measures how mixed or uncertain a group is at each decision node. If all the orders in a group are high risk, entropy is zero because there is no uncertainty. If the group is split evenly between high risk and low risk, entropy is at its highest. At each step the tree picks the feature and split point that reduces entropy the most. This reduction is called information gain. Using entropy felt

more meaningful for this business problem because it gives a clearer picture of how well each split is actually separating risky orders from safe ones.

Decision Tree Structure

The chart below shows the full structure of the trained tree. Each internal node shows a question the model is asking about a feature, and each leaf at the bottom shows a final prediction. The color of each node indicates which class is dominant at that point, and the numbers show the entropy and sample counts. Reading the tree from top to bottom gives a clear picture of the logic the model is using to classify each purchase order.

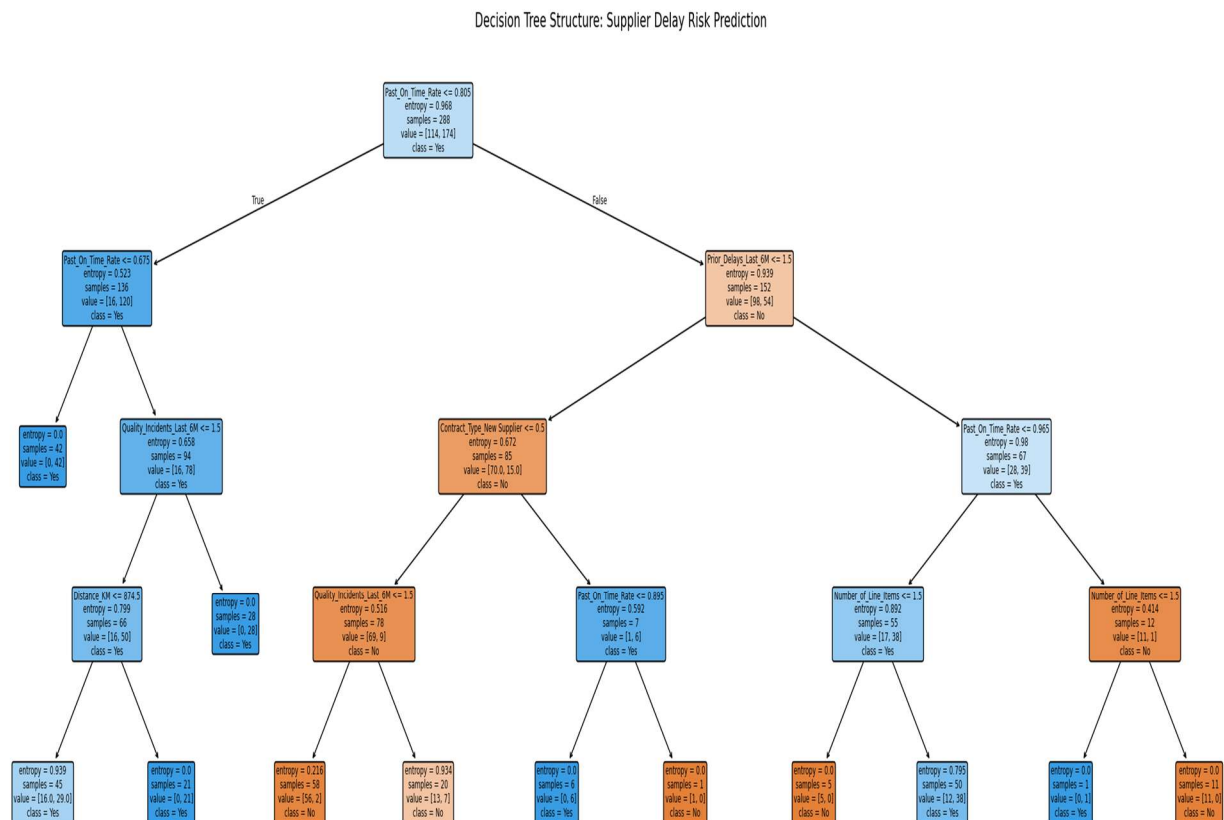


Figure 2: The trained Decision Tree structure with max depth of 4

4. Model Evaluation

Confusion Matrix

The confusion matrix below breaks down the predictions into four categories: true positives, true negatives, false positives, and false negatives. The rows represent the actual labels and the columns represent what the model predicted. Looking at this chart tells me not just how many predictions were right overall but also where the model is making mistakes and which type of error is more common.

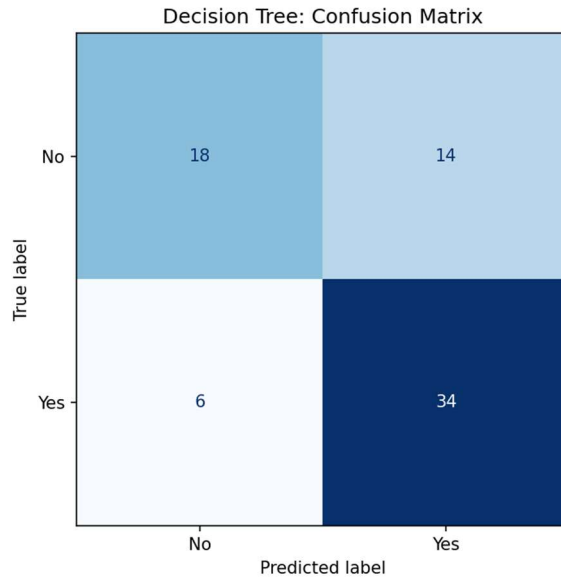


Figure 3: Confusion matrix showing correct and incorrect predictions on the test set

A false positive in this context means the model predicted an order was high risk but it actually arrived on time. This results in some wasted effort from the procurement team following up on an order that did not need attention. A false negative is the more serious mistake. It means the model predicted an order was safe but a delay actually occurred. Because no one followed up, the business had no chance to intervene before the problem reached the client.

Evaluation Metrics

The table below summarizes the key evaluation metrics measured on the test set.

Metric	Score
Training Accuracy	0.8715
Testing Accuracy	0.7222
Precision	0.7083
Recall	0.8500
F1 Score	0.7727

Training vs Testing Accuracy and Overfitting

The chart below compares the accuracy on the training data with the accuracy on the testing data. The training accuracy is 0.8715 and the testing accuracy is 0.7222. The gap between the two is about 0.15 which suggests some degree of overfitting even with the depth limit applied. The model learned the training patterns quite well but did not fully transfer all of that performance to new unseen data. This is a known tradeoff with Decision Trees. Limiting max depth to 4 helped reduce overfitting compared to an unconstrained tree, but some gap remained.

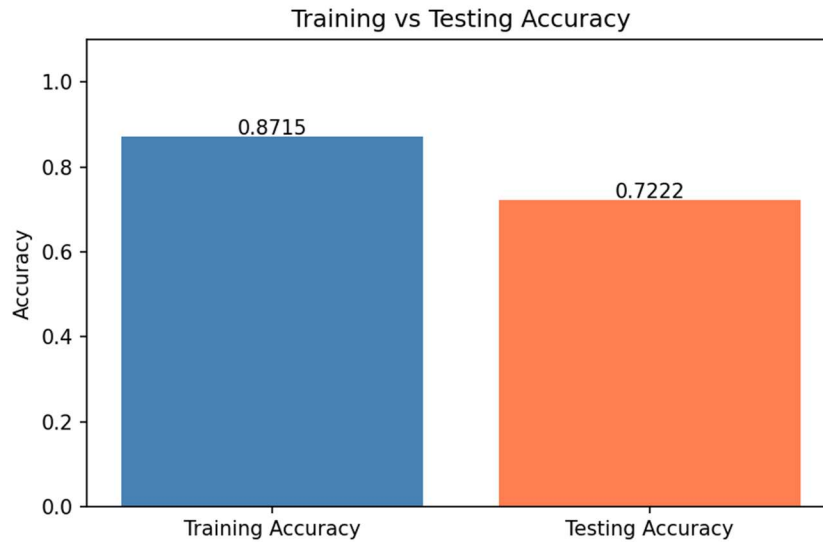


Figure 4: Training accuracy vs testing accuracy comparison

For this business problem, recall is the most important metric. The recall score of 0.85 means the model caught 85 percent of all actual high risk orders in the test set, which is a strong result for a procurement risk tool. Missing a high risk order is a costlier mistake than flagging a low risk order for extra attention, so maximizing recall is the right priority here.

5. Feature Importance

After measuring how well the model performed, I wanted to understand which features it relied on the most when making its predictions. During training, the Decision Tree evaluated many possible splits and kept track of which features were most useful for separating high risk orders from low risk ones. Features used more often and that led to cleaner separations received higher importance scores.

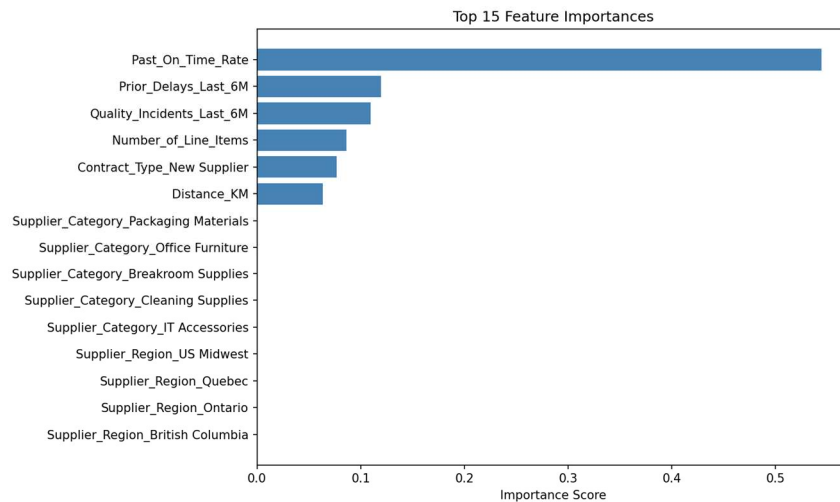


Figure 5: Top 15 most important features from the trained Decision Tree

The chart shows that `Past_On_Time_Rate` is by far the most important feature with an importance score of 0.544. This means the model relied on a supplier's historical delivery rate more than any other piece of information when making its predictions. `Prior_Delays_Last_6M` and `Quality_Incidents_Last_6M` follow as the next most important features, capturing recent supplier behavior patterns. `Number_of_Line_Items` and `Contract_Type_New Supplier` also appeared in the top five, suggesting that order complexity and new supplier relationships carry meaningful risk.

These results make intuitive sense from a procurement perspective. A supplier with a poor on time history is more likely to delay future orders. A supplier who has had recent quality problems or frequent delays in the past six months is showing active warning signs. And a new supplier with no established track record introduces uncertainty that experienced suppliers do not have.

6. Business Insights and Limitations

Key Business Insights

Based on what the model learned, the procurement team can take the following practical steps.

- Building a supplier scoring system based on `Past_On_Time_Rate` and `Prior_Delays_Last_6M` would let the team rank suppliers by reliability and avoid routing critical orders to low scoring ones.
- Orders placed with new suppliers should automatically receive extra monitoring since there is no established track record to rely on.
- `Quality_Incidents_Last_6M` signals that a supplier is actively struggling. Orders from suppliers with recent quality problems could be given shorter follow-up intervals or assigned backup options in advance.
- The model gives the procurement team a prioritized list of high risk orders so they can focus their limited time on the orders that actually need attention rather than reviewing everything manually.

One Limitation of the Model

The most important limitation to acknowledge is that this dataset was generated artificially rather than collected from real procurement operations. Real delays can be caused by events like weather disruptions, factory closures, port congestion, or supply chain shortages that are difficult to simulate in a generated dataset. Additionally, the `Backorder_History` column had 127 missing values which I filled with the most frequent value. If those missing entries were actually a meaningful signal, filling them this way might have introduced some noise into the model.

Why Human Judgment Still Matters

The Decision Tree model is a useful decision support tool but it should never replace human judgment entirely. The model learned from historical data and it cannot account for things happening right now that were not captured in the training records. A supplier may have recently improved its operations. A specific client may be unusually important to the business even if the order looks low risk on paper. The best way to use this model is to let it do the sorting and let the procurement team do the deciding. The model highlights where to look. The team applies their knowledge, experience, and current context to decide what to actually do about it.